

STAT 312: HW Assignment 4 - Unit 06

Instructor: Dr. Md Mutasim Billah

Submission Guidelines

Please follow these guidelines carefully when completing and submitting your homework. Clear and well-organized work helps ensure fair and accurate grading.

- There are **5 problems**.
- Submit all homework assignments through **Canvas**.
- If you prefer to work with **pen and pencil**:
 - Download and/or print the homework assignment.
 - Scan your completed work and combine all pages into a **single PDF file**.
 - Upload the PDF to the corresponding homework assignment on Canvas.
- If you prefer to use a **writing device** or type your solutions, you may complete the homework electronically and submit your final work directly as a **PDF**.
- You may also need to submit an **R file** if the homework includes a programming task.
- Your work must be **clear, organized, and legible**. Please follow these presentation guidelines:
 - Clearly circle or otherwise mark your final answers so they can be easily located on the page.
 - Show all necessary work for **nontrivial calculations**. Answers without sufficient supporting work may receive little or no credit.
- Name your submitted PDF using the following format:
`firstname_lastname_HW#.pdf`
For example: `john_smith_HW5.pdf`

Problem 4.1

Find the following probabilities if $Z \sim N(\mu = 0, \sigma = 1)$.

Instruction: You are required to use the **Z-table provided in class**. If you did not collect a copy, you can download it from the **Unit 06 Activities** section of the course website. Do **not** use a calculator to compute probabilities from z -values.

You are also required to **draw an appropriate sketch for every part of this problem**.

(a) $P(Z < -0.68)$

(b) $P(Z > 2)$

(c) $P(0 < Z < 1.73)$

Problem 4.2

Find the indicated value of z if $Z \sim N(\mu = 0, \sigma = 1)$.

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You are also required to **draw an appropriate sketch for every part of this problem**.

(d) Find z if $P(Z > z) = 0.5$.

(e) Find z_0 if $P(Z < z_0) = 0.8643$.

(f) Find z if $P(-z < Z < z) = 0.90$.

(g) Find z if $P(-0.20 < Z < z) = 0.0673$.

(h) What is the 32nd percentile of $Z \sim N(\mu = 0, \sigma = 1)$?

(i) What is the 68th percentile of $Z \sim N(\mu = 0, \sigma = 1)$?

Problem 4.3

Instruction: You are required to use the **Z-table provided in class**. If you did not collect a copy, you can download it from the **Unit 06 Activities** section of the course website. Do **not** use a calculator to compute z -values from probabilities. You may also **draw an appropriate sketch for every part of this problem**, if that helps.

Biologists are studying the weights of 11-year-old American boys. This population of weights is normally distributed with a mean of $\mu = 81$ pounds and a standard deviation of $\sigma = 9$ pounds.

Let $Y =$ **the weight of a randomly selected 11-year-old American boy**.

- (a) What is the probability that the weight of a randomly selected boy is less than 91.44 pounds?

- (b) What is the probability that the weight of a randomly selected boy is greater than 85.86 pounds?

- (c) What is the probability that the weight of a randomly selected boy is between 74.61 pounds and 88.38 pounds?

- (d) The probability that the weight of a randomly selected boy is greater than k pounds is 0.0250. What is k ?

- (e) What is the 78th percentile among the weights?

Problem 4.4

One method of arriving at economic forecasts is to use a consensus approach. A forecast is obtained from each of a large number of analysts; the average of these individual forecasts is the consensus forecast. Suppose that the individual 1996 January prime interest-rate forecasts of all economic analysts are approximately normally distributed with mean 7% and standard deviation 2.6%.

If a single analyst is randomly selected from among this group, what is the probability that the analyst's forecast of the prime interest rate will:

Hint: If we define $Y = \text{interest-rate forecast}$, what is the distribution of Y ?

(a) Exceed 11%?

(b) Be less than 9%?

Problem 4.5

Achievement test scores of all high school seniors in a state have mean 60 and variance 64. A random sample of $n = 100$ students from one large high school had a mean score of 58.

Hint: Calculate the probability that the sample mean is at most 58, given n .

(a) Is there evidence to suggest that this high school is inferior? Find the necessary probability and comment.

(b) If we had $n = 10$ students, could we answer part (a), given the information in this problem? If yes, why? If no, why? Either way, explain your reasoning.